

Adapttime: Method Overview

Thesis project

Current protocol

Abstract

Adapttime studies whether a small fitted adaptor can combine a time-series foundation model with fixed, calendar-aligned retrieval while preserving the official TIME test protocol. This note defines the current univariate `full_ridge_shared` formulation; it reports no empirical result.

1 Chronological protocol

Each dataset and forecast term is divided into a fixed datastore, adaptation training, adaptation validation, and the untouched official TIME test interval. The first three regions end strictly before the test. Ridge candidates are fit once on adaptation training, K and α are selected on adaptation validation, and the selected coefficients are frozen for test. There is no train-plus-validation refit.

Let H be the forecast horizon, T_{test} TIME’s configured test length, and $n = \lfloor T_{\text{test}}/H \rfloor$ its official complete test-window count per variate. Adaptation training contains $2n$ window origins and validation contains n , giving three times the test-window count before test. Their interval lengths are $H + (2n - 1)s_f$ and $H + (n - 1)s_f$, where the prime stride s_f depends on frequency rather than on H : 127 for intraday/hourly, 27 for business-daily/daily, 11 for weekly, 5 for monthly, and 3 for quarterly data. The remaining earlier observations form the datastore.

The ridge history length is the same as in the inherited vanilla TIME run for its backbone: 8192 for Chronos-2, 4096 for TS-ICL, and 2048 for Chronos-Bolt. TS-RAG retains its separate native length of 512 when used as an external control.

For a query time t , let d_{max} be the fixed datastore endpoint and P the dataset period. The latest eligible neighbor date is

$$d_t = \min(t - H, d_{\text{max}}) - (\min(t - H, d_{\text{max}}) - t) \bmod P. \quad (1)$$

Earlier candidates walk backward from d_t by a datastore stride that is a multiple of P . This keeps the datastore fixed while aligning its phase to the query.

By default every eligible earlier datastore date is accessible. An optional global window cap is divided evenly across variates, rounded down, and retains the most recent dates; every variate must retain at least P datastore dates. If the complete adaptation-window plan, requested foundation context, or this minimum datastore cannot be supplied, no ridge is fitted: the task evaluates TIME’s unchanged official test windows with vanilla forecasts and records the vanilla-only fallback reason.

Let q_t be the finite fraction of a query history. The default gate requires $q_t \geq 0.8$ before retrieval. Datastore histories satisfy the same threshold, and their complete retrieved future must be finite. A query is RAG-eligible only when it has at least K_{max} valid chronological neighbors, where $K_{\text{max}} = 15$ for the default grid. Every candidate K uses this common query support and consumes the first K neighbors from the same ordered list; fewer than K_{max} valid neighbors makes the query ineligible for all candidates.

2 Full shared ridge

For each query, V is the vanilla foundation forecast and C is the forecast obtained when the K retrieved neighbor trajectories are supplied as covariates. For neighbor i , Y_i is its observed horizon and N_i is the foundation model’s vanilla forecast from that neighbor’s own history. The full design is

$$X = [V, C, Y_1, \dots, Y_K, N_1, \dots, N_K]. \quad (2)$$

The paired Y_i and N_i terms expose neighbor residual information. Let $s_{t,c}$ be the root mean squared seasonal difference over the complete history before query t for channel c . Only pairs whose two original dates are finite contribute, and their count is the denominator. One no-intercept ridge vector is shared across all forecast horizons and fitted by MSSE:

$$\hat{\beta}_\alpha = \arg \min_{\beta} \sum_{t,c,h} \left(\frac{(Y - V)_{t,c,h} - (X\beta)_{t,c,h}}{s_{t,c}} \right)^2 + \alpha \|\beta\|_2^2, \quad (3)$$

$$V' = V + X\hat{\beta}_\alpha. \quad (4)$$

The primary setting is $K = 10$, $\alpha = 10^{-2}$. Validation searches $K \in \{1, 5, 10, 15\}$ and $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$. Delta, convex, per-horizon, and multivariate Adaption variants are outside the current protocol.

Ridge fitting uses complete adaptation-training rows only; it never applies an element-wise NaN mean to the normal equations. During validation and test, an ineligible query receives the vanilla output for both C and V' . This gate uses only query and datastore information available at forecast time. Missing future labels affect metric support only and cannot select the prediction path.

3 Evaluation

The official test compares Seasonal Naive, vanilla V , retrieval-context C at the selected K , and frozen Adaption V' . Raw MSE, MAE, MASE, and MSSE are retained as diagnostics. Performance uses scaled MASE: within each task, a method's equal-user MASE is divided by the matching Seasonal Naive MASE, then the task ratios are combined with a geometric mean. Missing dates are never removed or shifted: forecast errors skip non-finite target dates, while MASE and MSSE scales count only finite lagged pairs at their original timestamps. Conclusions require completed and inspected cluster artifacts.

Eligibility arrays, reason codes, effective training counts, and test RAG coverage are retained with the task artifacts. The resulting estimand is a transparent gated hybrid: Adaption on eligible windows and vanilla otherwise. An insufficient-history task is the limiting case with zero eligible Adaption windows and vanilla predictions for every learned-method branch.